

Double dumbbell — Solution

X21 (2021) — English translation

A1

0.30 pts

Find the angular velocity vector of the second rod at the initial moment of time. Express your answer in terms of $\vec{\omega}$.

At the moment the rotation begins, the ball at the end of the second rod is motionless. Therefore, the rods begin to rotate in opposite directions. Since the balls are at rest and the lengths of the rods are the same

$$\vec{\omega}_2 = -\vec{\omega}_1$$

A2

0.20 pts

At the initial moment of time, find the angular velocity vector $\vec{\omega}_0$ of the second rod in the reference system associated with the first rod. Express your answer using $\vec{\omega}$

From the law of addition of angular velocities

$$\vec{\omega} = \vec{\omega}_2 - \vec{\omega}_1$$

Комбинируя equality с resultom parta A1

$$\vec{\omega} = -2\vec{\omega}$$

A3

0.50 pts

Find the potential energy of the load in the field of centrifugal force at a distance r from the axis of rotation accurate to an arbitrary constant. Express your answer using m , ω , r .

The potential energy can be obtained as the potential energy of a spring with a spring constant of $-m\omega^2$. Hence

$$W_p = -\frac{m\omega^2 r^2}{2} + C$$

In what follows, we will assume that the constant is equal to zero.

A4

0.50 pts

Let the second rod rotate at an angle φ relative to the first. Find its angular velocity ω in this position. Express your answer using ω and φ .

As mentioned earlier, the Coriolis force does not change the kinetic energy of the system. Then the law of conservation of mechanical energy is satisfied in the system

$$W_k + W_p = E$$

В initial moment времени

$$2m\omega^2 L^2 = W_k$$

$$-2m\omega^2 L^2 = W_p$$

Thus

$$E = 0$$

В произвольный момент времени

$$W_k = \frac{m\omega^2 L^2}{2}$$

$$W_p = -2m\omega^2 L^2 \cos^2\left(\frac{\varphi}{2}\right)$$

Hence

$$\vec{\omega} = -2\vec{\omega} \cos\left(\frac{\varphi}{2}\right)$$

A5

0.50 pts

Find the speed of the load in a fixed frame of reference. Express your answer in terms of ω , L , φ .

One velocity component will be directed along the second rod, and the other will be perpendicular to it. We will receive

$$v_1 = \omega L \sin(\varphi)$$

$$v_2 = \omega L(1 + \cos(\varphi)) - \omega L$$

For квадрата скорости we get

$$v^2 = v_1^2 + v_2^2 = (2\omega L)^2 \left(\left(\cos^2\left(\frac{\varphi}{2}\right) - \cos\left(\frac{\varphi}{2}\right) \right)^2 + \sin^2\left(\frac{\varphi}{2}\right) \cos^2\left(\frac{\varphi}{2}\right) \right)$$

Ненце после простых преобсований

$$v = 2\omega L \sqrt{2 \left(\cos^2\left(\frac{\varphi}{2}\right) - \cos^3\left(\frac{\varphi}{2}\right) \right)}$$

B1

0.40 pts

Let us denote the tension force of the rods as T

Find ускорения грузов a_{1l}, a_{1p} в проекции на ось OX and acceleration ball/spherenupра a_2 в проекции на ось OY . Answers выtimesandme via/through m, g, T and φ .

$$a_{1x} = \frac{T \cos(\varphi)}{m}$$

$$a_{1x} = -\frac{T \cos(\varphi)}{m}$$

$$a_{2y} = -\left(\frac{T \sin(\varphi)}{m} + g \right)$$

B2

0.60 pts

In the indicated position, find the speed of the weights v_1 and the speed of the hinge v_2 .

From the law of conservation of mechanical energy

$$\frac{2m}{2}(v_1^2 + v_2^2) = 2m \left(\frac{v_0^2}{2} - gL \sin(\varphi) \right)$$

Equation кинематической связи

$$v_1 = v_2 \tan(\varphi)$$

Hence

$$v_1 = \sqrt{v_0^2 - 2gL \sin(\varphi)} \sin(\varphi)$$
$$v_2 = \sqrt{v_0^2 - 2gL \sin(\varphi)} \cos(\varphi)$$

B3

1.00 pts

Write down the equation of motion of the hinge relative to any of the masses, connecting v_1 , v_2 , a_1 , a_2 , L and φ

The hinge moves in a circle relative to the loads. Hence

$$\frac{v^2}{L} = a_{1x} \cos(\varphi) - a_{2y} \sin(\varphi)$$

Since скорости ball/sphere и груза перпендикулярны

$$v^2 = v_1^2 + v_2^2$$

and finally

$$\frac{v_1^2 + v_2^2}{L} = a_{1x} \cos(\varphi) - a_{2y} \sin(\varphi)$$

B4

0.70 pts

Find the tension force T from the equation obtained in step B3. Express the answer in terms of m , v_0 , g , L , φ

Combining the results of the first three points

$$\left(\frac{v_0^2}{L} - 2g \sin(\varphi) \right) = \frac{T}{m} + g \sin(\varphi)$$

hence

$$T = m \left(\frac{v_0^2}{L} - 3g \sin(\varphi) \right)$$

B5

1.00 pts

Let us assume that at some moment the weights are lifted off the surface.

Find value угла φ_{neg} at this point in time.

Condition under which weights are lifted off the surface

$$T \sin(\varphi) = mg$$

Substituting answer на part B4

$$m \left(\frac{v_0^2}{L} - 3g \sin(\varphi) \right) \sin(\varphi) = mg$$

hence

$$3 \sin^2(\varphi) - \frac{v_0^2}{gL} \sin(\varphi) + 1 = 0$$

Let us find корни квадратного уравнения

$$\sin(\varphi) = \frac{\frac{v_0^2}{gL} \pm \sqrt{\left(\frac{v_0^2}{gL}\right)^2 - 12}}{6}$$

Первым достигается меньший корень, а потому окончательный answer

$$\sin(\varphi) = \frac{\frac{v_0^2}{gL} - \sqrt{\left(\frac{v_0^2}{gL}\right)^2 - 12}}{6}$$

B6

0.50 pts

Find the minimum value v_0 at which the weights will come off the surface and the value of the angle φ' at the moment of lifting

The discriminant of the quadratic equation must be non-negative, hence

$$v_0 \geq \sqrt{2\sqrt{3}gL}$$

Минимум достигается in the case of нулевого дискриминанта, therefore

$$\sin(\varphi_{max}) = \frac{1}{\sqrt{3}}$$

C1

0.20 pts

At what values of R does the collision of shells occur before their separation from the surface under the action of the rods? Express your answer using L, φ ,

Condition under which the collision occurs before the separation of

$$R \geq L \cos(\varphi)$$

C2

1.00 pts

Let immediately after the collision the projections of the speed of the load on the axes OX and OY be equal to v_{1x} and v_{1y} , respectively.

Express проекцию скорости ball/sphere на вертикальную ось v_{2y} via/through v_{1x}, v_{1y} and φ_0 .

The load moves in a circle relative to the hinge, so

$$\pm v_{1x} = (v_{1y} - v_{2y}) \tan(\varphi_0)$$

$$v_{2y} = v_{1y} \pm \frac{v_{1x}}{\tan(\varphi_0)}$$

C3

1.30 pts

Find the velocity of the hinge v_2 immediately after the impact. Express your answer using v_0, g, L, R

Let's solve the problem in the reference frame of the center of mass. Let v_x and v_y be the modulus of the relative speed of the loads and the modulus of the vertical velocity of the load relative to the hinge, respectively. Then the expression for kinetic energy in the reference frame of the center of mass is the following

$$\frac{mv_x^2}{4} + \frac{mv_y^2}{2} = W_k$$

После соударения это also выполняется.

From the equation кинематической связи

$$\frac{v_x}{v_y} = \operatorname{tg}(\varphi_0)$$

Ссу после соударения это coratio also выполняется. Therefore, from energy conservation it follows, что после соударения magnitudes относительных скоростей остаются постоянными. Это говорит о том, что грузы обмениваются друг с другом горизонтальными составляющими скоростей, а ball/spherенир обменивается с ними вертикальными составляющими скоростей. Thus, for/at соударении ball/spherенир останавливается.

$$v_2 = 0$$

C4

1.30 pts

Find the modulus of the load velocities v_1 immediately after the collision. Express the answer in terms of v_0, g, L, R

All kinetic energy goes to the loads. From the law of conservation of energy we get the answer

$$v_1 = \sqrt{v_0^2 - 2g\sqrt{L^2 - R^2}}$$